higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

ELECTROTECHNICS N4

15 NOVEMBER 2019

This marking guideline consists of 8 pages.

Copyright reserved

QUESTION 1

1.1 True
1.2 True
1.3 False
1.4 True
1.5 False
1.6 True
1.7 True
1.8 False
1.9 False
1.10 True

(10 × 1) **[10]**

QUESTION 2

2.1 2.1.1 Series:

$$V_T = V_1 + V_2 + V_3 \checkmark$$
$$I_T = \frac{V_T}{R_T} \checkmark$$
$$R_T = R_1 + R_2 + R_3 \checkmark$$

Parallel:

$$V_T = I_T R_T \checkmark$$
$$I_T = I_1 + I_2 + I_3 \checkmark$$
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \checkmark$$

(6)

2.1.2 Total power:

$$P_T = V_T I_T \checkmark$$

$$P_T = \left(I_T\right)^2 R_T \checkmark$$

$$P_T = \frac{\left(V_T\right)^2}{R_T} \checkmark$$

(3)

2.2 First draw a circuit ABCDEFA

From circuit ABEFA:

$220 - 200 - 0,1I_1 - 9,8I_2 = 0$

$\therefore 20 - 0,1I_1 - 9,8I_2 = 0 \ldots\ldots\ldots\

QUESTION 3

3.1 It states that when a conductor cuts or is cut by magnetic flux, an EMF is generated in that conductor and the magnitude of the generated EMF is proportional to the rate at which the conductor cuts or is cut by the magnetic flux. (3)

3.2 Given:

$C_1 = 5\mu F$

$C_2 = 10\mu F$

$C_3 = 15\mu F$

$V_T = 100V$

3.2.1

$$C_S = \frac{1}{\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}}$$

$$C_S = \frac{1}{\frac{1}{5}+\frac{1}{10}+\frac{1}{15}} \checkmark$$

$$C_S = 2,73\mu F \checkmark$$ (2)

3.2.2

$$Q_T = Q_1 = Q_2 = Q_3$$

$$VC = 100x2,73 = 273\mu F \checkmark$$

$$V = \frac{Q}{C}$$

$$V_1 = \frac{273}{5} = 54,6V \checkmark$$

$$V_2 = \frac{273}{10} = 27,3V \checkmark$$

$$V_3 = \frac{273}{15} = 18,2V \checkmark$$ (4)

3.3
- Separetely-excited
- Self-excited (2)

3.4 Given:

p = 4
c = 2 x 4 = 8
N = 350 r/p
E = 260 V
Φ = 0,05 Wb

$$E = \frac{2ZNp\phi}{60}$$

$$Z = \frac{60x260x8}{2x350x4x0,05}$$ ✓✓

$$Z = 891,4 = 891$$ ✓✓

$$Conductors / slot = \frac{Conductors}{Slots} = \frac{891}{120} = 7,43$$ ✓✓

= 8 conductors per slot ✓ (7)

[18]

QUESTION 4

4.1 4.1.1 The field excitation is supplied from an independent DC source such as a battery or separate DC generator.

4.1.2 The field windings are connected to the armature windings in various ways.

4.1.3 The field windings and load are all connected in parallel to one another, where I_{sh} = shunt field current.

4.1.4 The field windings and load are connected in series to one another, where I_{se} = series field current.

4.1.5 It is a generator which has both a series and shunt field windings in one circuit, which normally comes as either a short shunt or a long shunt.

(5 × 2) (10)

4.2 Given:
No, of turns = 100
A = 40 cm²
N = 1 500 r/m
Φ = 0,05 T

4.2.1 $F = \frac{N}{t} = \frac{1500}{60} = 25Hz$ ✓✓ (2)

4.2.2 $T = \frac{1}{f} = \frac{1}{25} = 0,04\sec$ ✓ (1)

4.2.3 $E_m = 2\pi BANn$

$E_m = 2\pi x0,05x40x10^{-4}x100x\frac{1500}{60}$ ✓✓

$E_m = 3,14V$ ✓ (3)

4.2.4 $e = E_m Sin\theta$

$e = 3,14Sin30^0$ ✓

$e = 1,57V$ ✓ (2)
[18]

QUESTION 5

5.1 5.1.1
- X_L will be greater than Xc, as X_L and Xc are in direct opposition.
- The resultant will be (X_L – Xc).
- The overall power factor (cos Φ) will be lagging.

5.1.2
- Vc will therefore be greater than X_L.
- The resultant will be (Xc – X_L).
- The overall power factor (Cos Φ) will be leading.

(2 × 3) (6)

5.2 5.2.1
- X_L will be smaller than Xc, as Xc and X_L are in direct opposition.
- The resultant will be (Xc – X_L).
- The overall power factor (Cos Φ) will be leading.

5.2.2
- Ic will therefore be smaller than X_L.
- The resultant will be (X_L – Xc).
- The overall power factor (Cos Φ) will be lagging.

Copyright reserved

(2 × 3) (6)

5.3 Given:
S = 25 kVA
N1 = 500 turns
V1 = 3 kV
f = 50 Hz

5.3.1

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

$$\therefore V_2 = \frac{V_1 N_2}{N_1} = \frac{3x10^3 x40}{500} = 240V$$ ✓✓

5.3.2 $S = VI$

$$\therefore I_1 = \frac{25x10^3}{3x10^3} = 8,33A$$ ✓

$$I_2 = \frac{25x10^3}{240} = 104,17A$$ ✓

5.3.3 $E = 4,44\phi_m fN$

$$\phi_m = \frac{3x10^3}{4,44x50x500} = 27mWb$$ ✓✓

(3 × 2) (6)
[18]

QUESTION 6

6.1
- It is relatively cheap and easy to transport the final product, which is electrical power.
- The pollution is kept away from densely populated areas. (2)

6.2
- Ring-feeder system
- Radial-feeder system (2)

6.3
- Eskom Generation
- Eskom Transmission
- Eskom Distribution (3)

6.4 6.4.1 SHORT SHUNT:

Given:

$R_v = 2000\Omega$

$R_A = 0,5\Omega$

$V = 10V$

$I = 0,05A$

$$R_{app} = \frac{V}{I} = \frac{10}{0,05} = 200\Omega \quad ✓$$

$$I_v = \frac{V}{R_m} = \frac{10}{2000} = 0,005A \quad ✓$$

$$R_x = \frac{V}{I - I_v} = \frac{10}{(0,05 - 0,005)} = 222,22\Omega \quad ✓$$

$$\%Error = \frac{R_x - R_{app}}{R_x} X100\%$$

$$= \frac{222,22 - 200}{222,22} X100\% \quad ✓$$

$$= 10\% \ ✓$$

LONG SHUNT:

Given:

$R_v = 2000\Omega$

$R_A = 0,5\Omega$

$V = 5V$

$I = 1A$

$$R_{app} = \frac{V}{I} = \frac{5}{1} = 5\Omega \quad ✓$$

$$R_x = \frac{V}{I} - R_A = \frac{5}{1} - 0,5 = 4,5\Omega \quad ✓$$

$$\%Error = \frac{R_x - R_{app}}{R_x} X100\%$$

$$= \frac{4,5 - 5}{4,5} X100\% \quad ✓$$

(9)

$= -11,11\%$ ✓

6.4.2 The minus sign indicates that the apparent resistance calculated, was too high. (2)
[18]

TOTAL: 100